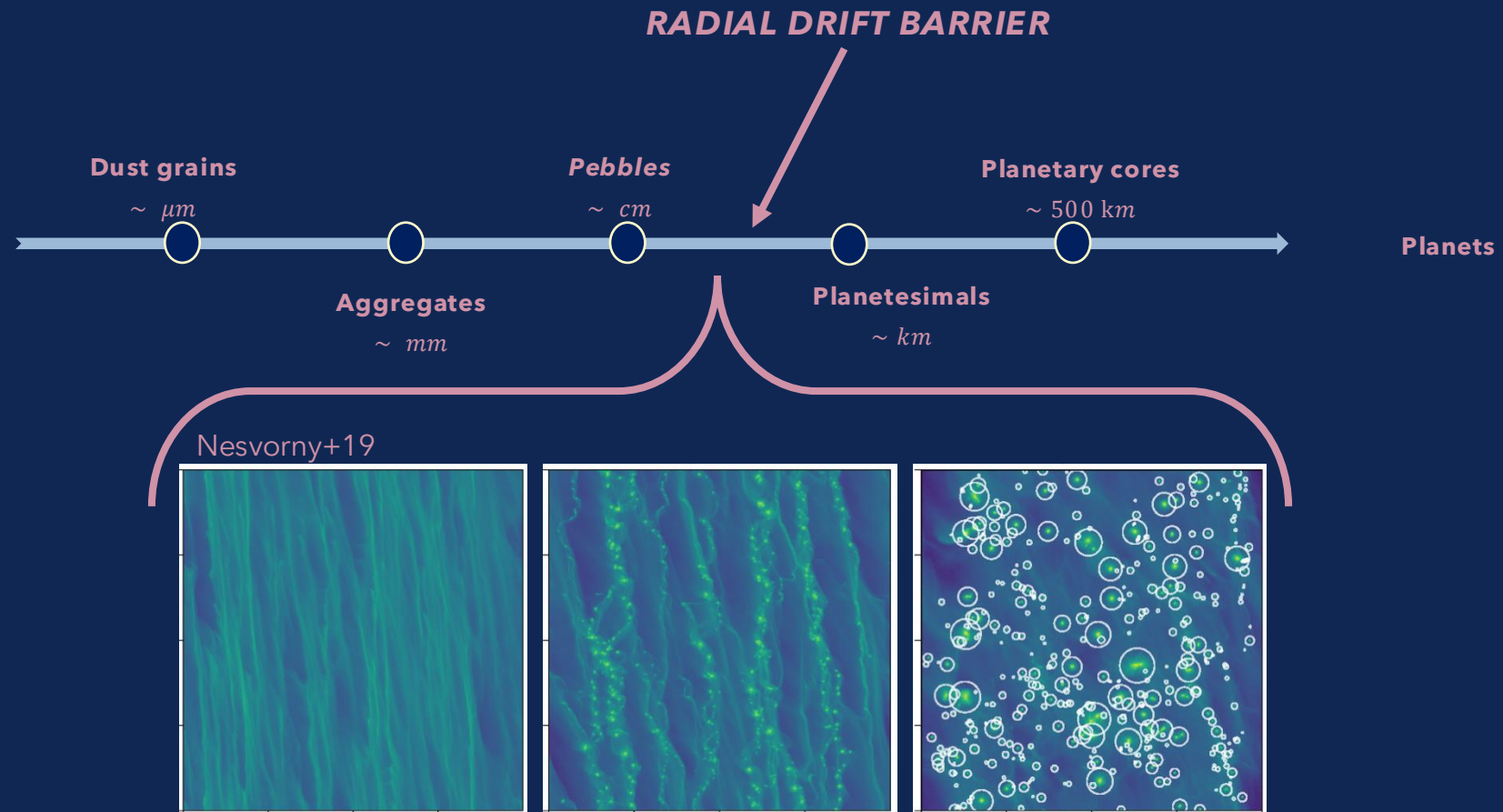


From dust filaments to observables: tracing streaming instability

Chiara Scardoni

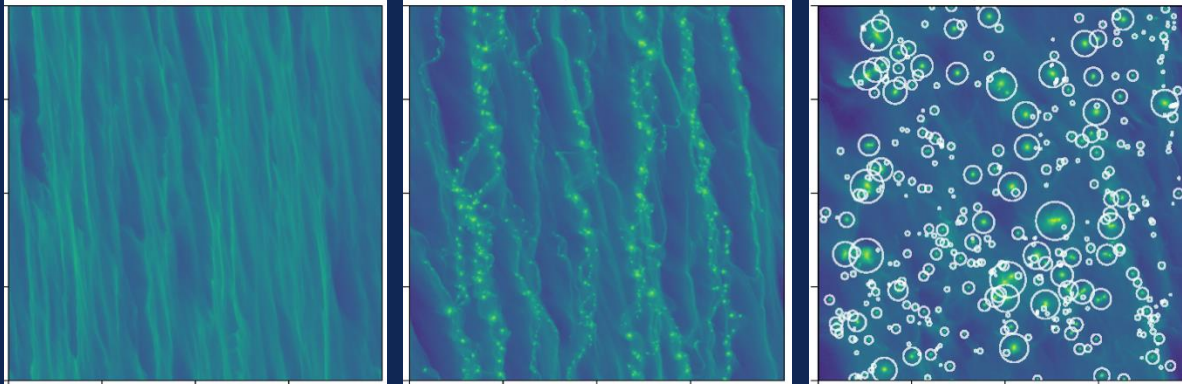
Collaborators: Booth R., Clarke C., Rosotti G., Ribas A, Villenave M., Zagaria F.

STREAMING INSTABILITY (SI)



THE OBSERVATIONAL CHALLENGE

Nesvorny+19

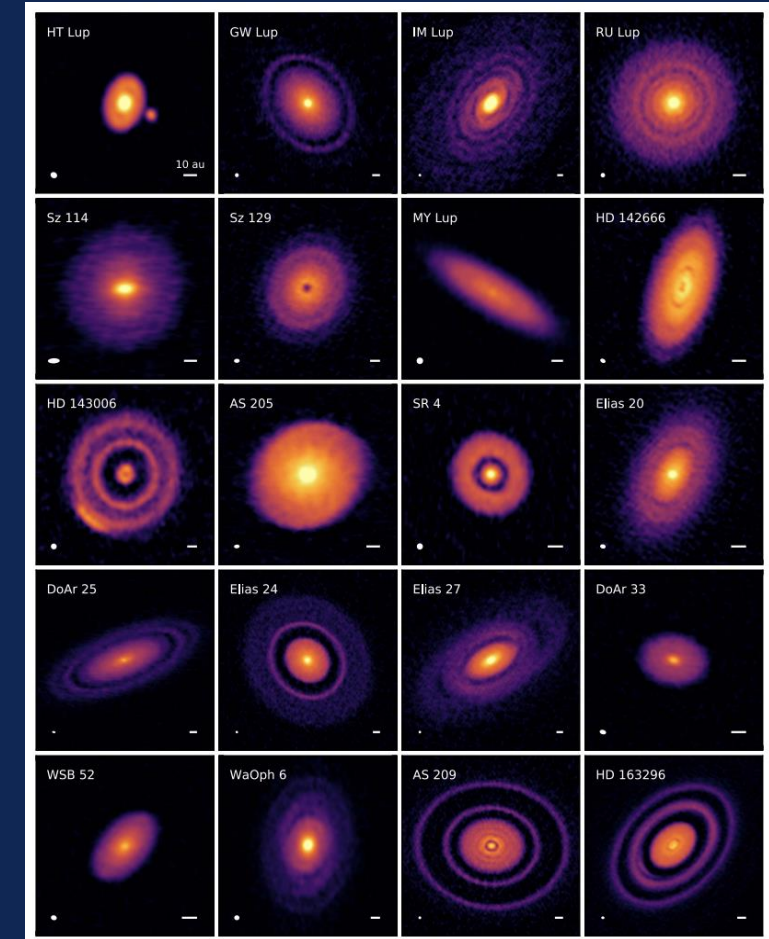


sub-au dusty substructures

Highest resolution observations: ~ 5 au beam size

→ **SI substructures cannot be observed directly**

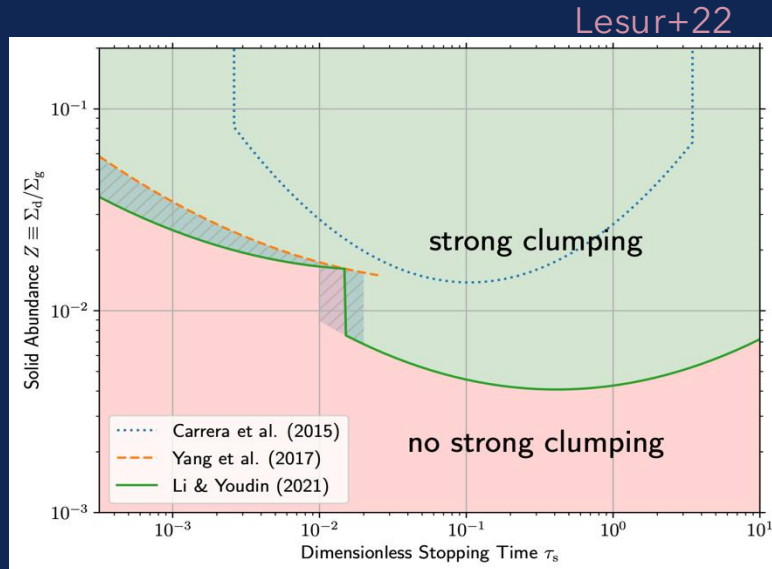
Andrews+18



INDIRECT OBSERVATIONS

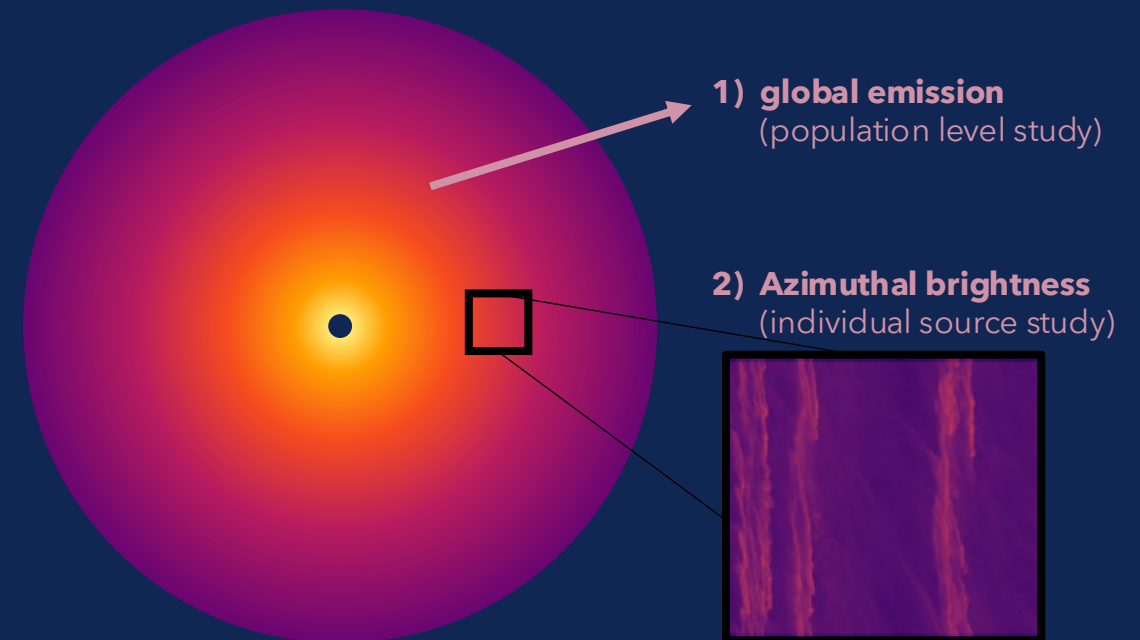
METHOD 1:

Are the disc's properties compatible with the conditions to trigger streaming instability? (Zagaria+23)



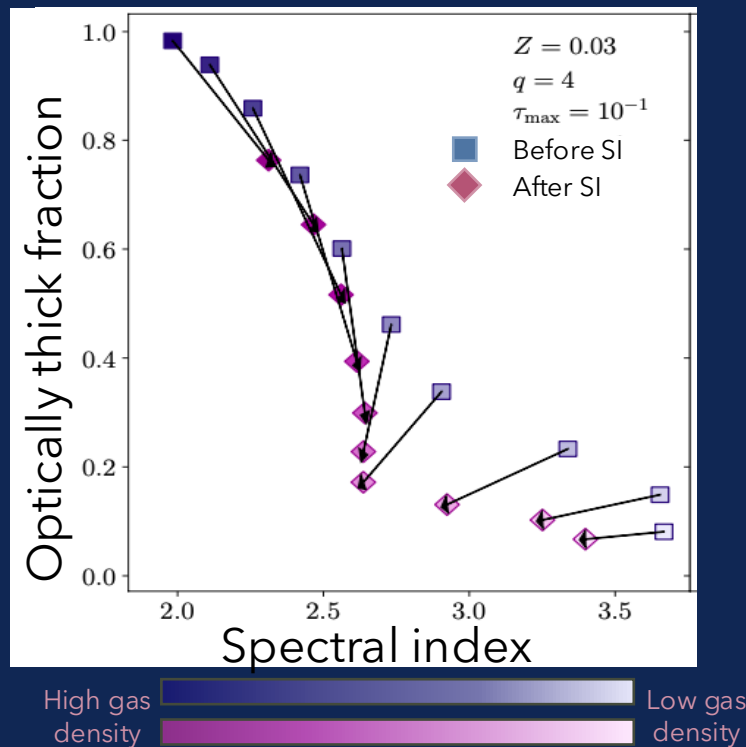
METHOD 2:

Are the SI-induced substructure leaving any signature in the disc's mm-emission? (Scardoni+21,24)

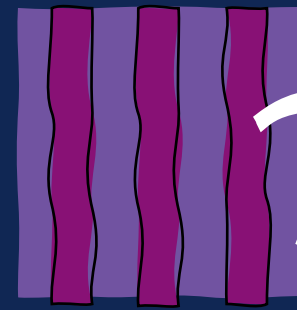


GLOBAL EMISSION

Perform SI simulation and analyse the total flux before/after SI action



Region undergoing SI



Optically thick filaments
 $\rightarrow$ big particles ($St \sim 1$)

Optically thin background
 $\rightarrow$ small particles ($St \ll 1$)

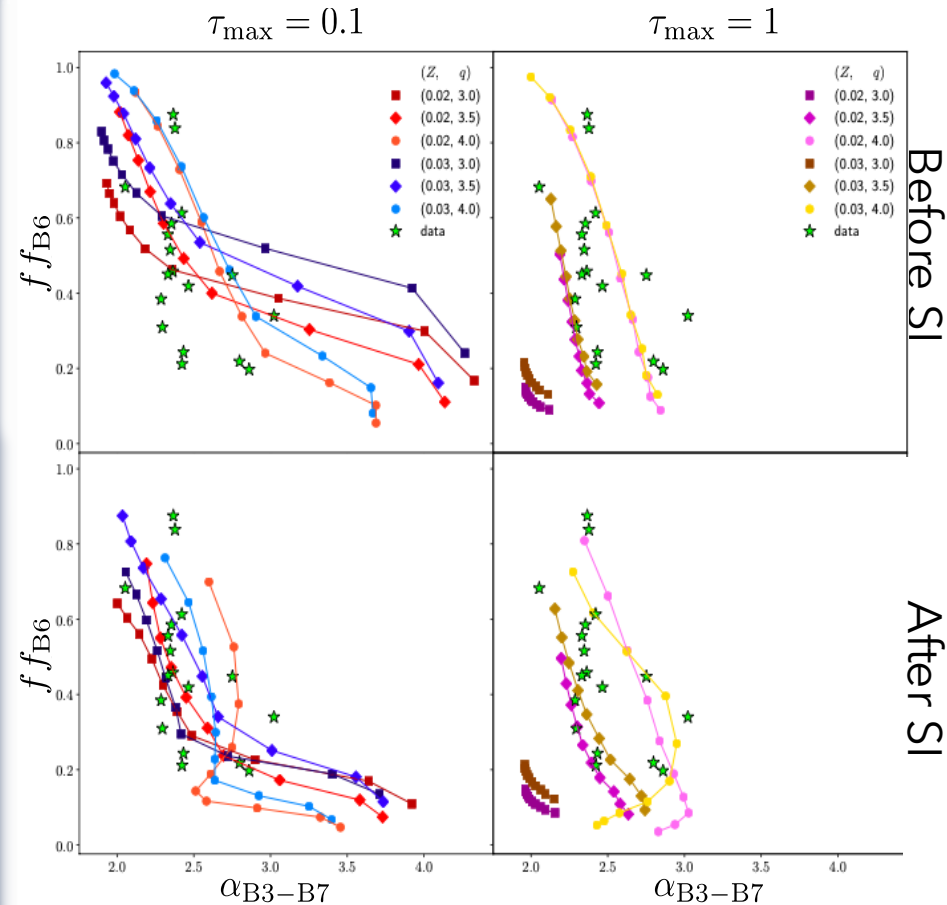
When a particle goes into the filament its emission is (partially) hidden

$\rightarrow$ the total flux is reduced after filament formation

After filament formation, the particles in the background drive the spectral index

GLOBAL EMISSION: RESULTS

Data by Tazzari+21

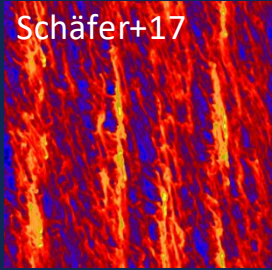


Effect of filament formation on the observable quantities:

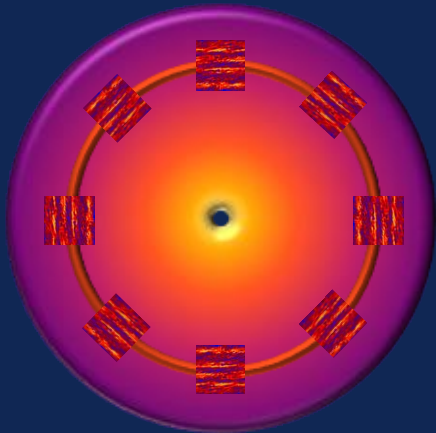
1. ff always declines after SI action
2. α either increase or decrease (depending on the optical properties of the particles going into the filaments)

SI drives the simulations towards the area of the plane occupied by the data

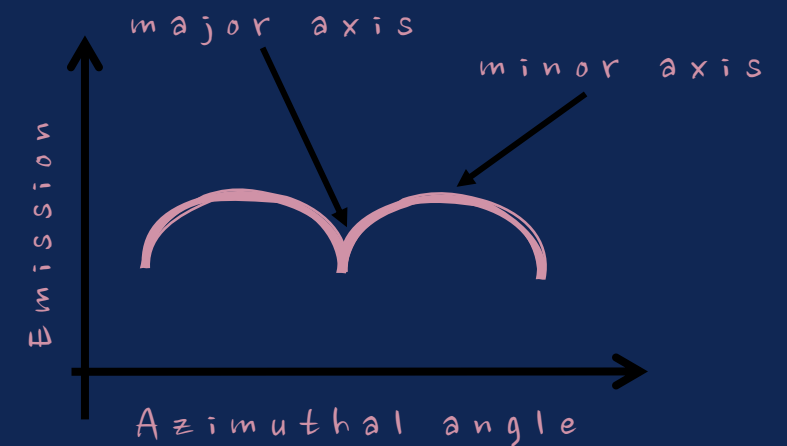
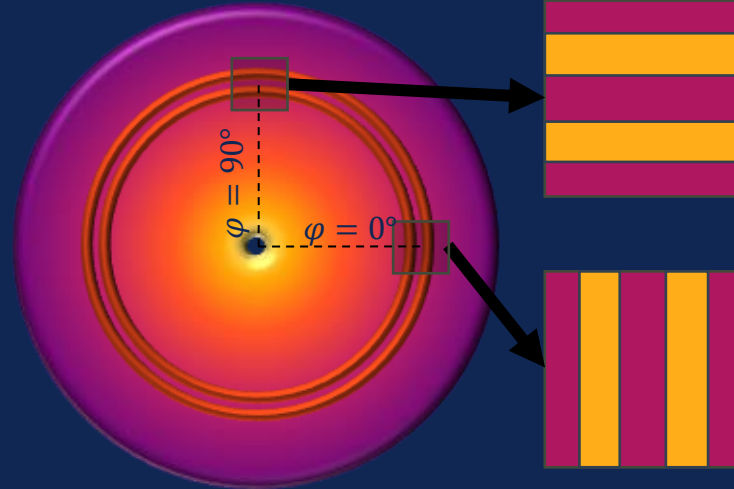
AZIMUTHAL BRIGHTNESS: MODEL



In the disc

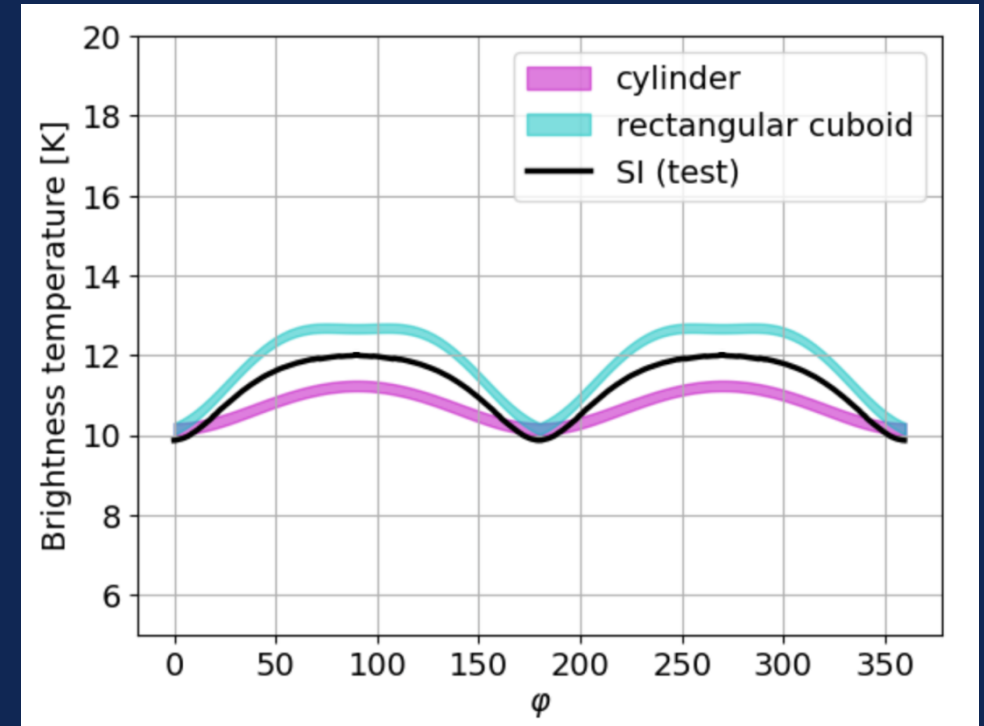
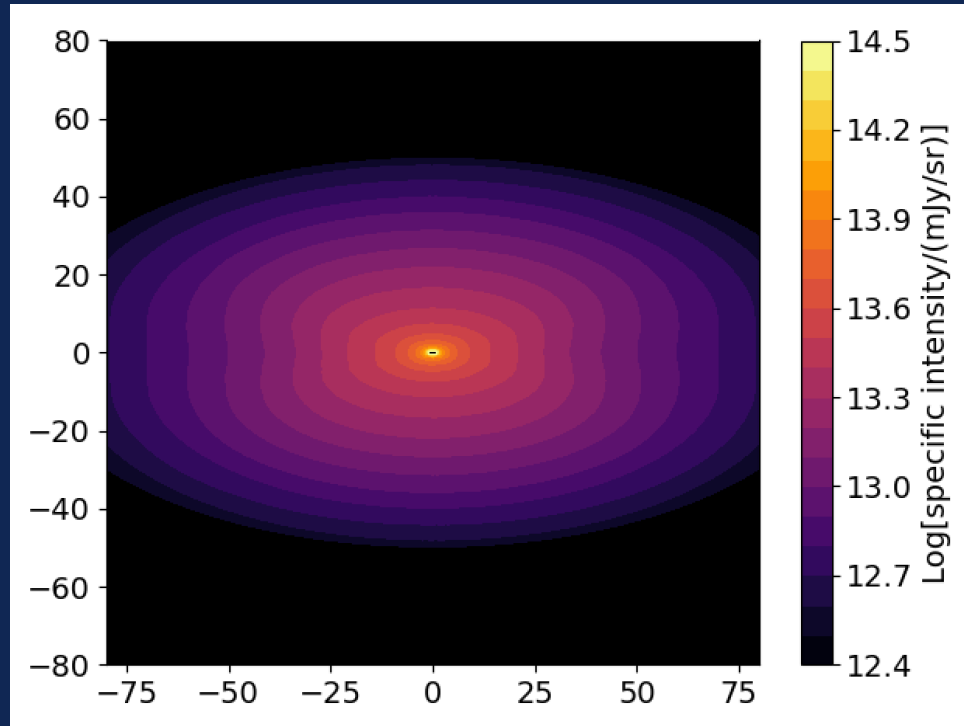


Emission pattern of unresolved optically thick ring-like substructures over an optically thin background



AZIMUTHAL BRIGHTNESS: SIMULATIONS

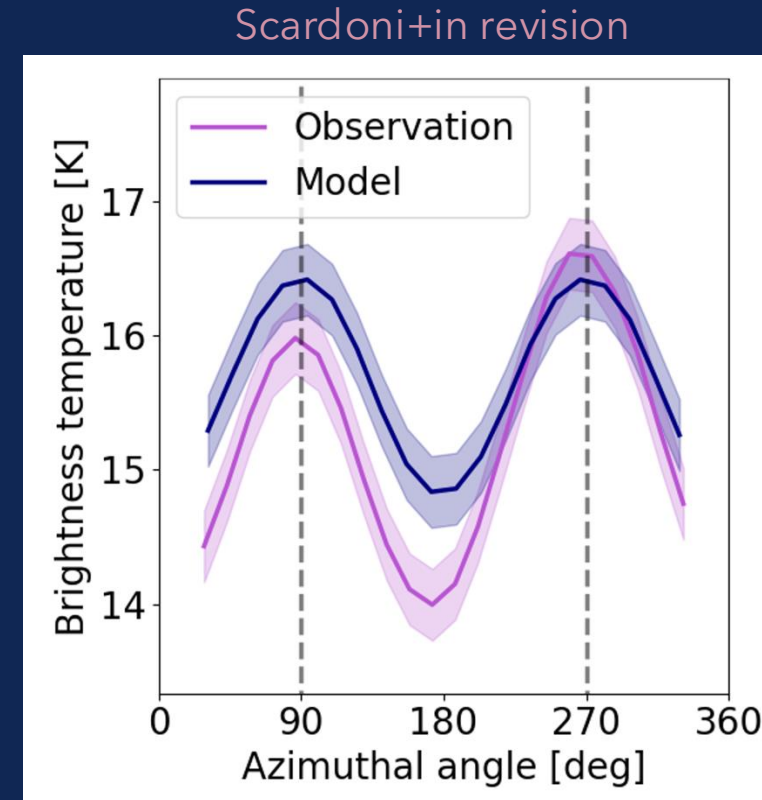
(assuming that SI is acting everywhere in the disc)



4 parameter analytical model
(rings' optical depth, background optical depth, rings' width and height)

SEARCH FOR THE SIGNATURE

- Search in the ALMA archive (inclined discs)
→ signature found in one source
- Model the disc (adding unresolved optically thick rings) and generate the synthetic observations (SIMIO, Kurtovich+25)
- We cannot fully discriminate the origin of these rings, but we can exclude some mechanisms
- Open the path to further search for this effect (e.g., other sources, other mechanisms)



SUMMARY & CONCLUSIONS

Study of indirect signature of SI-like ("ring-like") dusty dense substructures

	GLOBAL EMISSION	AZIMUTHAL SIGNATURE
THEORY	SI-like filaments reduce f_f and increase/decrease α	SI-like filaments cause higher (lower) emission along the minor (major) axis of inclined discs → effect modelled via a 4 parameter analytical model
DATA	SI-like filaments consistent with Lupus data (Tazzari+21)	Signature found in one disc

